UNSW SYDNEY

BENV7500 Programmable Cities – Assignment 3
Analysing the impact of urban configurations on wind speed around buildings in urban neighbourhoods from high-resolution numerical simulations

Sarfraj Shaik
zID – 5586793

Master of City Planning
School of Built Environment
UNSW Sydney

Table of Contents

List of Figures

1. Introduction: Problem and Rationale

Urban environments dramatically influence wind flow, with implications for pedestrian comfort, energy effiency, and pollution dispersion (T. R. Oke., et al., 2017; Britter and Hanna, 2003). The configuration of buildings determines how wind behaves at the street level — accelerating, decelerating, or forming turbulent eddies. This study seeks to quantify how different urban morphologies affect wind speed around buildings by identifying windward and leeward surfaces and calculating the wind speed differential.

Using data from high-resolution numerical simulations provided by the UrbanTALES dataset, we analyze the pedestrian-level wind field across various urban configurations. Understanding this relationship is vital for designing climate responsive cities and ensuring pedestrian safety and comfort in urban planning.

2. UrbanTALES Dataset and Input Files

The analysis utilizes the UrbanTALES dataset, an extensive resource of large-eddy simulation (LES) flow fields over idealized urban geometries (Nazarian., et al., 2023). Each simulation includes:

- _topo.txt: A 2D matrix of building heights representing urban topography, where non-zero values correspond to building surfaces.
- _ped.nc: A NetCDF file containing 2D wind speed data (Uped) at pedestrian level and corresponding x-y coordinates.

These datasets reflect diverse urban configurations — from uniformly spaced low-rise blocks to mixed-height and high-density layouts — and simulate wind flow from various directions (Blocken, 2015). The topography files define the physical layout, while the '.nc' files provide corresponding flow fields under a fixed inflow condition.

3. Methodology: Wind Flow Classification and Differential Analysis

This section explains the step-by-step process implemented in Python to analyse wind speed differentials across urban building arrays. The approach focuses on identifying windward and leeward surfaces based on topography and wind direction, then calculating and comparing wind speeds at those surfaces across multiple cases.

3.1 Reading and Preparing the Topography and Wind Data

The analysis begins with reading the topography data using a custom function. This function parses a .txt file containing the 2D array of building heights. Zero values represent fluid (air) cells, while non-zero values represent buildings (their heights). Wind direction is defined using a vector, such as [1,1], representing a diagonal wind at 45 degrees.

Using this vector, the script then identifies windward (front-facing) and leeward (rear-facing) surfaces by scanning the grid with a custom function 'find_fluid_front_and_rear_faces.'

3.2 Surface Classification

Each grid cell is classified into one of three categories — fluid, windward, and leeward — using integer labels (0,1,2). These labels are then assigned to the wind speed dataset (from '.nc' files), and mapped to textual labels. The result is a labelled Dataframe (df) where each cell contains its grid indices, wind speed (Uped), and surface classification.

3.3 Filtering and Statistical Analysis

The Dataframe is filtered to retain only windward and leeward cells. This filtered data is then used to compute the average wind speed for each surface type. The key metric, average wind speed difference, is then calculated as difference of mean leeward wind speed values from those of mean windward wind speed values.

3.4 Handling Missing Values

A recurring challenge was the presence of NaN values in Uped, especially for leeward cells. The code included a check to verify the absence of data. And to streamline future checks, a custom module 'nan_checker' was implemented. It included a function to inspect NaNs across categories. This ensures that missing data is identified before statistical calculations are performed.

3.5 Batch Processing Across Configurations

Using 'metadata-idealized.csv', the analysis is looped over multiple simulation cases with 0 degrees and 45 degrees wind directions. For each case:

- Topography and wind data are loaded
- Windward/leeward faces are identified
- Wind speed statistics are computed
- Results are appended to a summary table.

This master dataset contains average wind speeds at windward and leeward faces across multiple urban layouts and wind directions, enabling downstream comparative analysis.

4. Results

4.1 Correlation with Average Wind Speed Difference (Uped_diff)

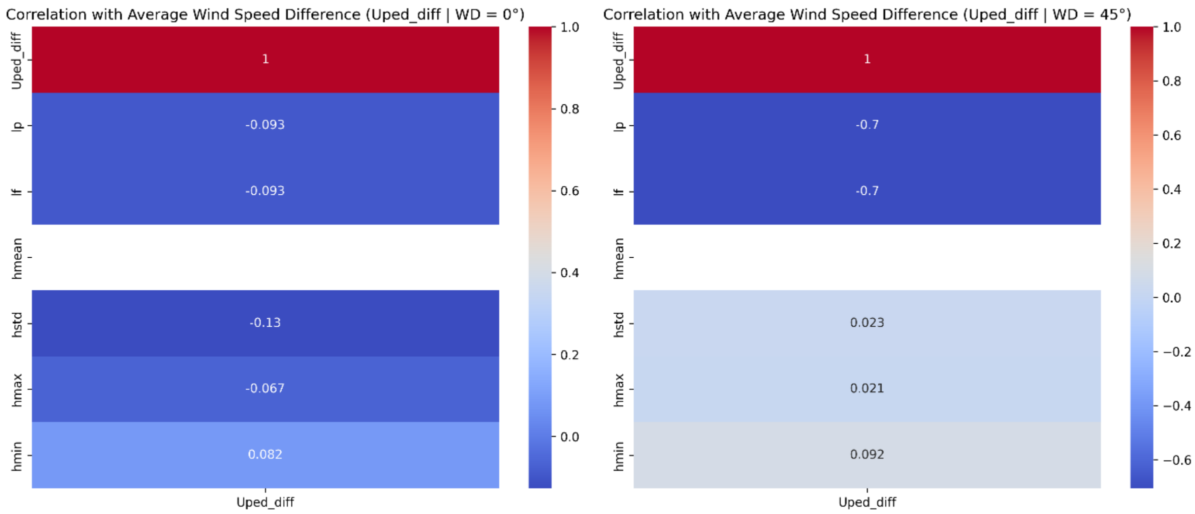

Figure 1 Correlation with Average Wind Speed Difference (Uped_diff)

This heatmap displays the Pearson correlation coefficients between 'Uped_diff' and various urban form parameters (Plan Area Density (lp), Frontal Area Density (lf),Building Height Mean (hmean),Standard Deviation in Building Height (hstd),Maximum Building Height (hmax), Minimum Building Height (hmin)) for wind directions of 0 and 45 degrees.

- At 0 degrees, the correlations are weak across all parameters. The highest is a slight negative correlation with hstd (-0.13), implying some weak relationship between height variability and wind speed difference.
- At 45 degrees, strong negative correlations emerge with lp (plan area density) and lf (frontal area density), both at -0.7, suggesting that more densely packed or exposed layouts decrease the difference between windward and leeward wind speeds.

Wind direction plays a substantial role in modulating the impact of urban form. Te 45 degrees wind direction seems more sensitive to density metrics.

4.2 Maximum Building Height (hmax) vs Average Wind Speed Difference (Uped_diff)

Both trends are nearly flat. Very tall buildings do not clearly increase or decrease wind speed. Contrary to expectation, maximum building height (hmax) alone is not a good predictor of wind sheltering.

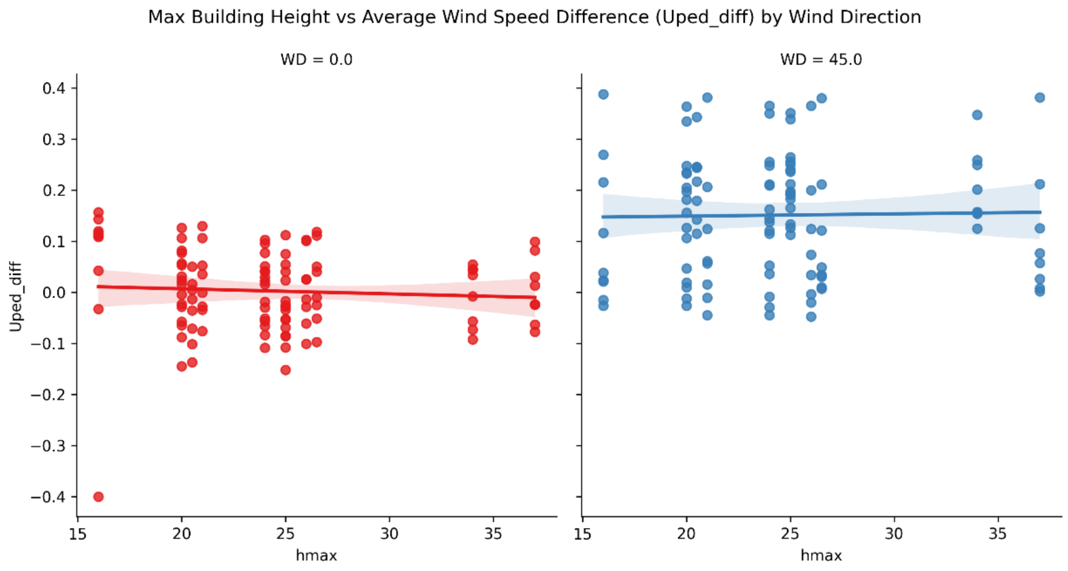

Figure 2 Maximum Building Height (hmax) vs Average Wind Speed Difference (Uped_diff)

4.3 Minimum Building Height (hmin) vs Average Wind Speed Difference (Uped_diff)

A slight upward trend is seen, more strongly under 45 degrees, suggesting that areas with higher minimum building heights (i.e., fewer shorter structures) experience higher 'Uped_diff.' Uniformly tall building bases might contribute to more pronounced windward-leeward contrasts.

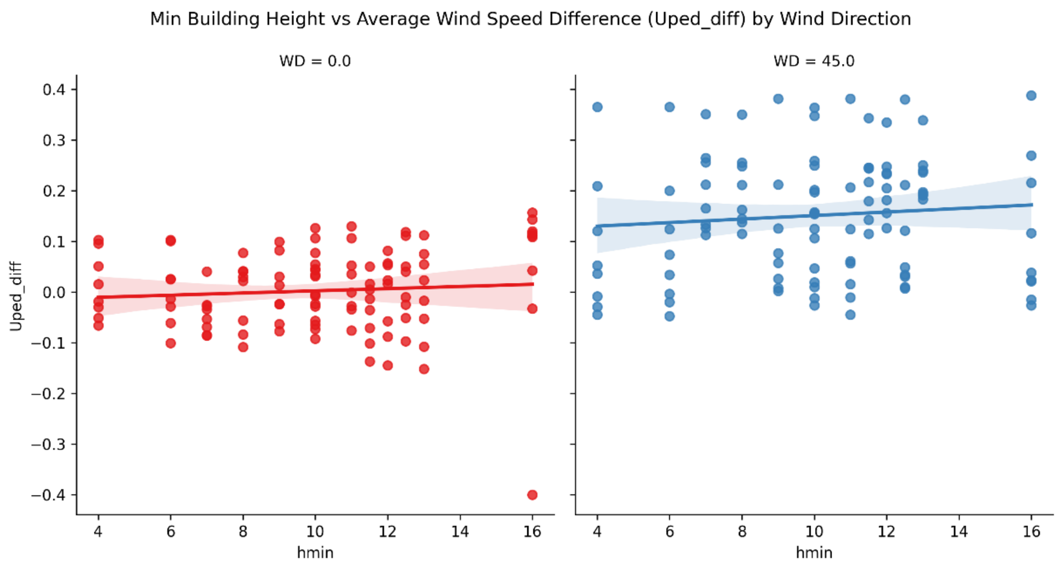

Figure 3 Minimum Building Height (hmin) vs Average Wind Speed Difference (Uped_diff)

4.4 Standard Deviation in Building Height vs Wind Speed Difference

Both plots are almost flat. A slightly decreasing trend under 0 degrees suggests that variation in height does not help wind protection and may actually reduce the leeward effect. Height heterogeneity (hstd) doesn't strengthen the winf speed gradient, and may introduce mixing/turbulence.

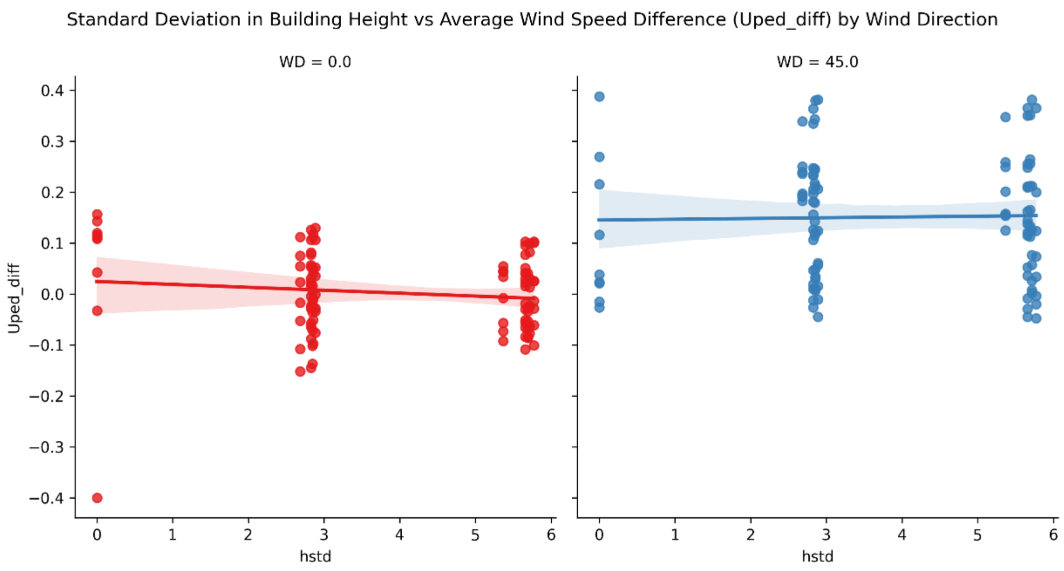

Figure 4 Standard Deviation in Building Height vs Wind Speed Difference

4.5 Effect of Building Height Metrics on Wind Speed Difference

This grid of scatter plots shows how 'Uped_diff' varies with 'hmax,hmin, and hstd' under both wind directions.

- Under 0 degrees, trends are flat or mildly negative for 'hmax' and 'hstd'
- Under 45 degrees, trends are flatter but slightly positive for 'hmin'

Minimum height (hmin) appears to slightly increase 'Uped_diff' under 45 degrees wind, suggesting that low-profile buildings may contribute to higher contrasts in wind exposure.

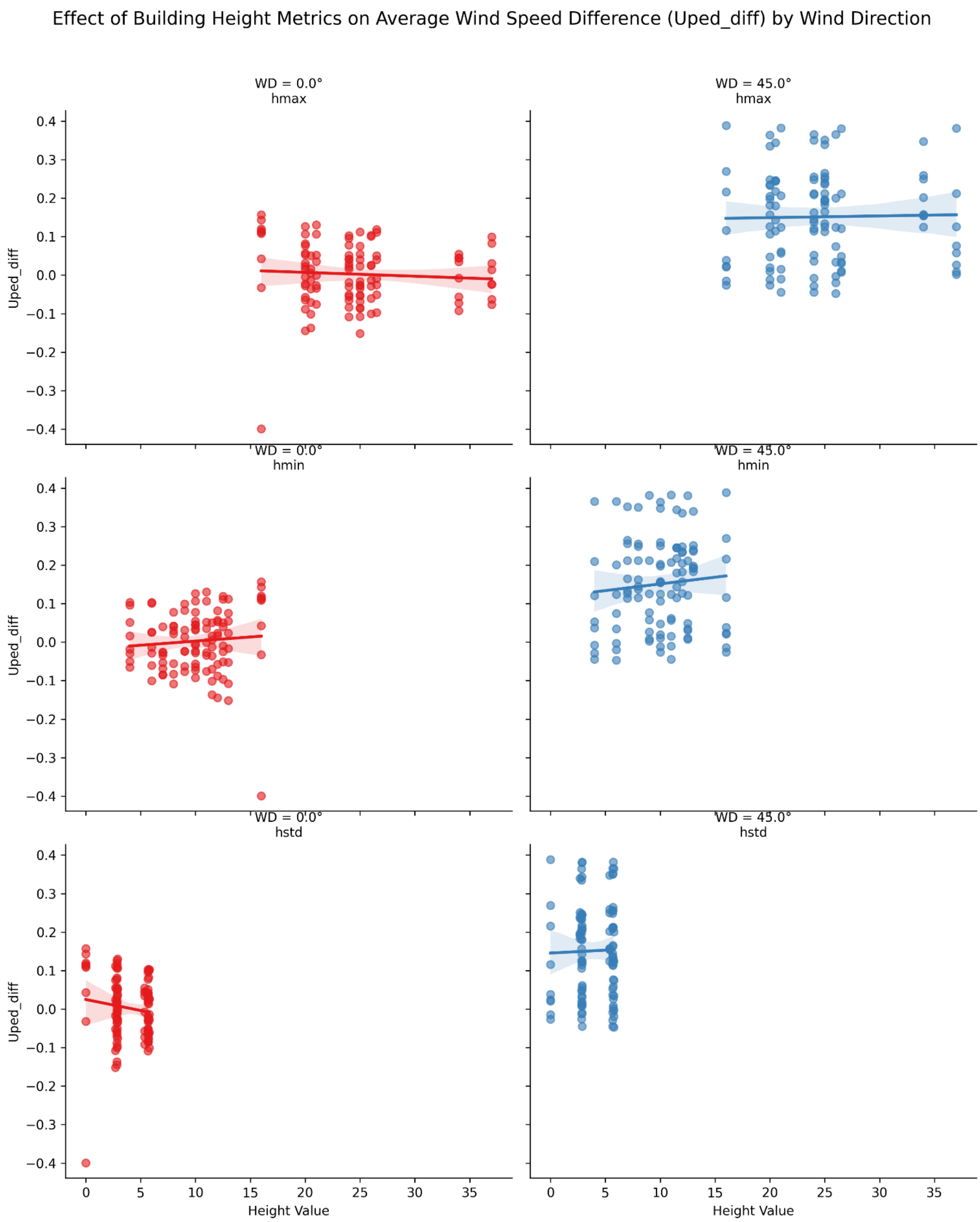

Figure 5 Effect of Building Height Metrics on Wind Speed Difference

4.6 Overall Effect of Plan Area Density (lp), Frontal Area Density (lf), and Maximum Height (hmax) on Average Wind Speed Difference (Uped_diff)

Bubble plots show ‘Uped_diff’ across combinations of ‘lp’ (x-axis) and ‘lf’(y-axis), with color denoting ‘Uped_diff’ and size representing ‘hmax’

- Higher densities (lp and lf) correlate with lower ‘Uped_diff’ (bluer tones)
- In aligned settlement cases, the trend remains consistent, but in staggered layouts, higher densities do not lead to greater wind shielding.

Aligned settlements enhance leeward wind protection more predictably than staggered ones.

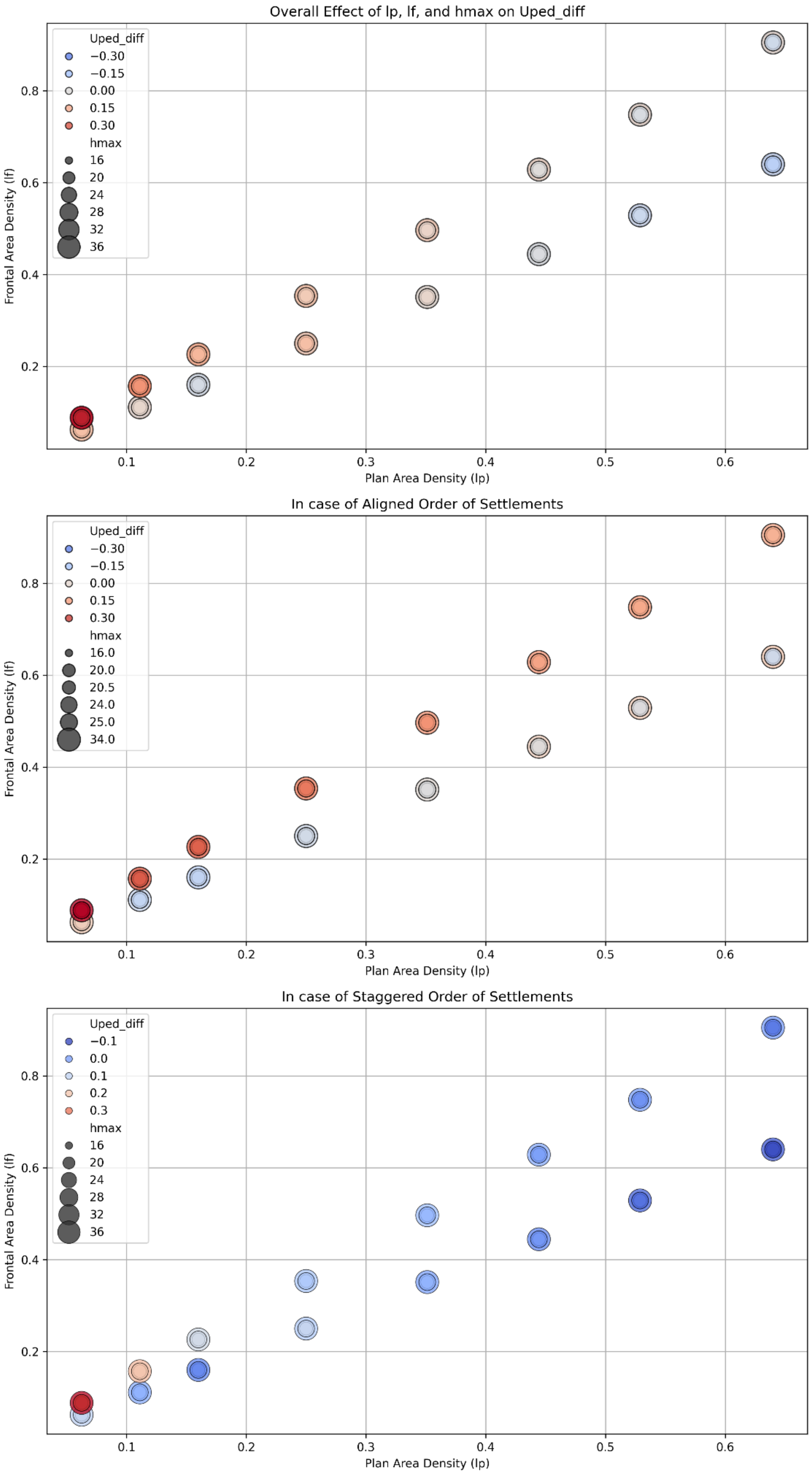

Figure 6 Overall Effect of Plan Area Density (lp), Frontal Area Density (lf), and Maximum Height (hmax) on Average Wind Speed Difference (Uped_diff)

4.7 Plan vs Frontal Density by Wind Direction, Building Heights, and Layouts

Split into subplots by wind direction and building height level, this chart shows the effect of Plan Area Density (lp) and Frontal Area Density (lf) combinations on Average Wind Speed (Uped_diff), grouped by settlement orientation (Aligned or Staggered)

- Under high-density and high-height scenarios, 'Uped_diff' reduces.
- Staggered patterns show a much weaker shelterinig effect, even at similar density levels.

Settlement orientation is a crucial determinant of pedestrian-level wind exposure — aligned rows create more pronounced wind shadows than staggered ones.

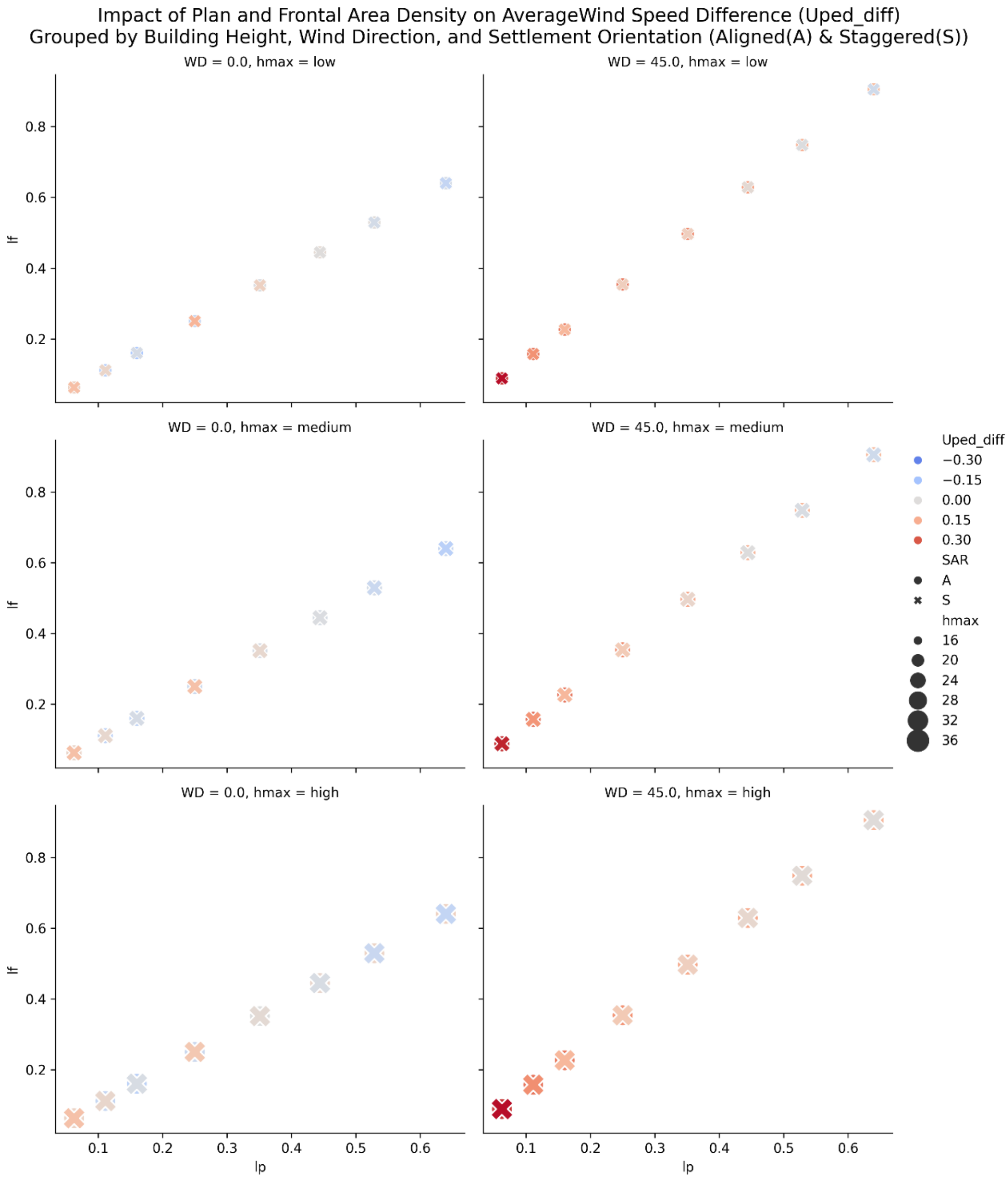

Figure 7 Plan vs Frontal Density by Wind Direction, Building Heights, and Layouts

4.8 Pairwise Relationships of Urban Parameters and Wind Speed Difference

This scatter matrix plot gives a holistic view of interactions between variables.

- Strong correlations between Plan Area Density (lp) and Frontal Area Density (lf), as expected.
- 'Uped_diff' has diffuse distribution with no clear clustering, except mild directional color separation (red for 0 degrees and blue for 45 degrees)

Urban density metrics are more interrelated than they are individually predictive of 'Uped_diff.'

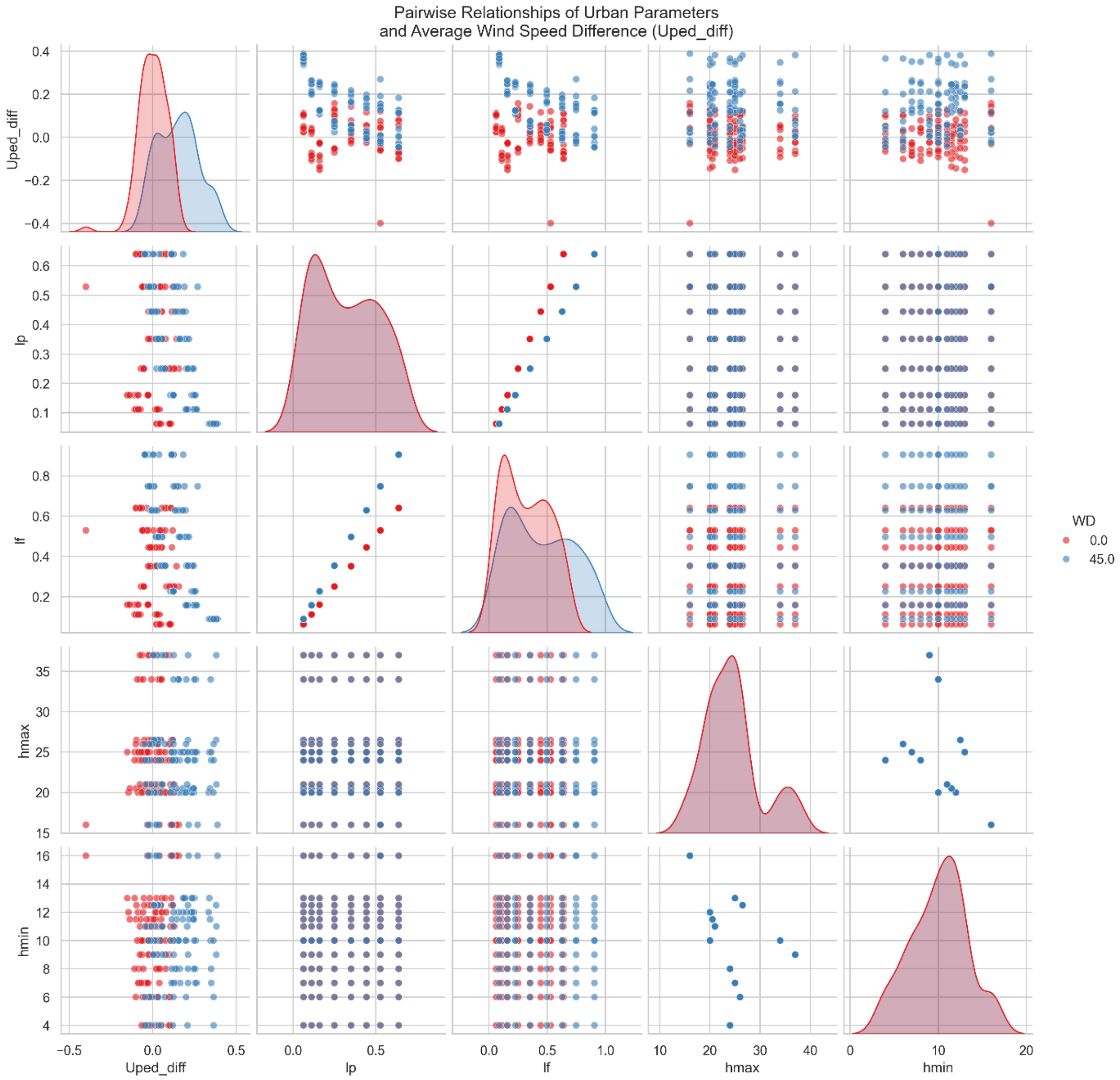

Figure 8 Pairwise Relationships of Urban Parameters and Wind Speed Difference

4.9 Critical Conclusion

This study offers compelling evidence that urban form — particularly building density, layout regularity, and minimum height — plays a more significant role in shaping pedestrian-level wind speed differences than maximum height or standard deviation in height.

- Plan and frontal area density are strong predictors of wind sheltering, especially under oblique wind (45 degrees)
- Aligned layouts systematically outperform staggered ones in creating effective wind shadows.
- Height uniformity appears more influential than height extremes.

5. Applications

The analysis framework developed in this study can be readily applied or extended in several domains.

5.1 Urban Design and Planning

The approach provides a computational basis to assess wind performance of proposed urban layouts before construction. By evaluating the impact of building arrangements and densities on pedestrian-level wind exposure, city planners can optimize for both ventilation and comfort. For instance, aligned grid patterns with moderately consistent building heights were shown to improve wind shadowing effects, which may be desirable in cold or windy climates.

5.2 Climate-Responsive Architecture

Architects and designers can use the findings to support passive design strategies (T. R. Oke., 2017), such as strategic placement of taller buildings or buffer zones based on prevailing wind directions. The ability to predict average wind speed difference (Uped_diff) allows for better façade orientation and massing in energy-conscious developments.

5.3 Environmental and Health Modelling

Understanding how building configurations modulate wind can enhance models of pollutant dispersion, heat stress, or airborne pathogen transmission in cities. UrbanTALES-style simulations can be integrated into larger-scale urban microclimate models or decision-support tools for resilience planning.

5.4 Policy and Regulation

The quantitative metrics explored (e.g., plan area density, frontal area density) can inform evidence-based urban policies and design codes. Rather than replying on empirical guidelines alone, planners could adopt these parameters as part of performace-driven zoning regulations or design competitions.

6. Limitations and Next Steps

6.1 Limitations

- Static Wind Conditions: Simulations assume steady-state flow from fixed wind directions (0 and 45 degrees). In reality, wind varies over time and interacts with thermal gradients, diurnal cycles, and synoptic conditions.
- Missing Data Challenges: The presence of NaN values in some cases complicated early analysis. While handled later with custom functions, it still introduces minor uncertainities in average calculations.
- Limited Variable Interactions: Some urban metrics (e.g., hstd) showed negligible correlation with 'Uped_diff', but their influence may become evident under nonlinear or coupled flow regimes, which were beyond this study's scope.
-

6.2 Next Steps

- Apply to Real Urban Data: Future work should replicate this pipeline on high-fidelity urban GIS or LiDAR models, where variability is more representative of actual cities.
- Incorporate Thermal Effects: Coupling wind simulations with heat flux and surface temperature would allow modelling of urban heat island mitigation strategies and pedestrian thermal comfort.
- Wind-Tunnel or Field Validation: Physical validation using scaled models in wind tunnels or in-situ measurements in cities could reinforce simulation outcomes.
- Expand Wind Scenarios: Adding more wind angles, varying speeds, and unsteady flow inputs (e.g., gusts) would enhance generalizability.

7. References

Blocken, B., 2015. *Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations,* s.l.: Building and Environment.

Jiachen Lu, N. N. M. A. H. E. S. K. A. M., 2023. *Novel Geometric Parameters for Assessing Flow Over Realistic Versus Idealized Urban Arrays,* s.l.: Journal of Advances in Modelling Earth System.

R. E. Britter., S. R. H., 2003. *FLOW AND DISPERSION IN URBAN AREAS,* s.l.: Annual Reviews.

T. R. Oke., G. M. A. C. A. V., 2017. *Urban Climates,* s.l.: Cambridge University Press.